

NVIDIA

TPUs

<https://docs.cloud.google.com/tpu/docs/system-architecture-tpu-vm/>

Control/ISA: Rather than using a general-purpose CPU in the accelerator, TPU V1 exposed a small set of high-level neural network instructions like *Read_host_Memory*, *Read_Weights*, *MatrixMultiply/Convolve*, *Activate*, and *Write_Host_Memory*, effectively a domain-specific ISA.

Memory architecture: Model weights weren't hardwired/"Burned in" and were instead stored in external 8 GiB DRAM weight memory and staged through a weight FIFO into the matrix unit. High potential for a memory bandwidth bottleneck, as TPU V1 was.

TPU performance depends heavily on mapping the workload onto the machine. Google's XLA stack tiles matrix operations to match the physical MXUs and chooses layouts compatible with the memory hierarchy.

Eric's opinion: I think we could explore tiled matrix engines plus vector/reduction/state-processing hardware with explicitly managed local SRAM and device memory. The compiler/runtime would lower the known Qwen graph into coarse accelerator commands and tile data around the available SRAM. The controller should probably stay independent of this interface with an x86 host, a simple hardware controller, or in the FPGA, the RISC-V softcore should all theoretically be able to submit the same command stream, which should let us compare RISC-V against a smaller BPU-specific command processor/ISA

DMatrix

NVIDIA + Groq

Etched

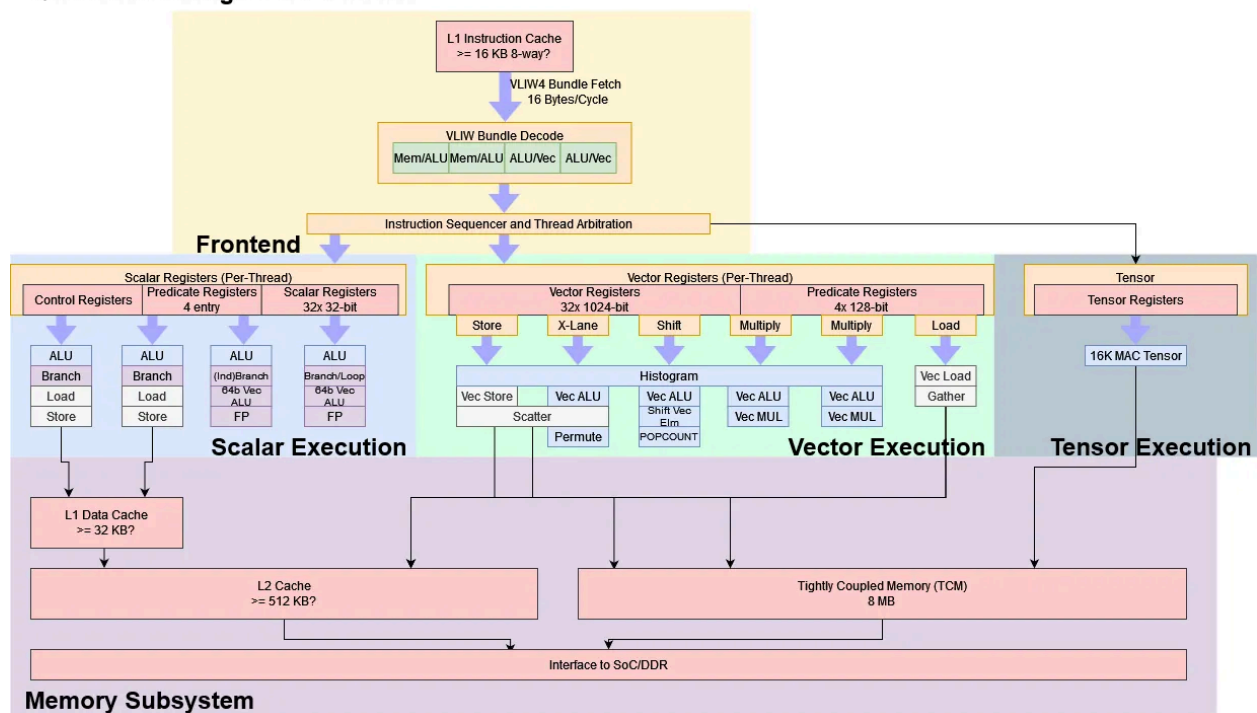
AMD

Qualcomm

Hexagon Neural Processing Unit (NPU) -

Hexagon runs scalar, vector and tensor processors in parallel, and can support virtual memory and caching (additionally can run compiled C). From the diagram, each “section” has their own dedicated registers:

Qualcomm Hexagon v73 + Tensor



Hexagon broad specs:

- Very Long Instruction Word approach (16 bytes wide), each ‘bundle’ can contain up to 4 instructions
- relies on a central shared memory (TCM)
- Uses 6-way simultaneous multithreading
- Branch prediction

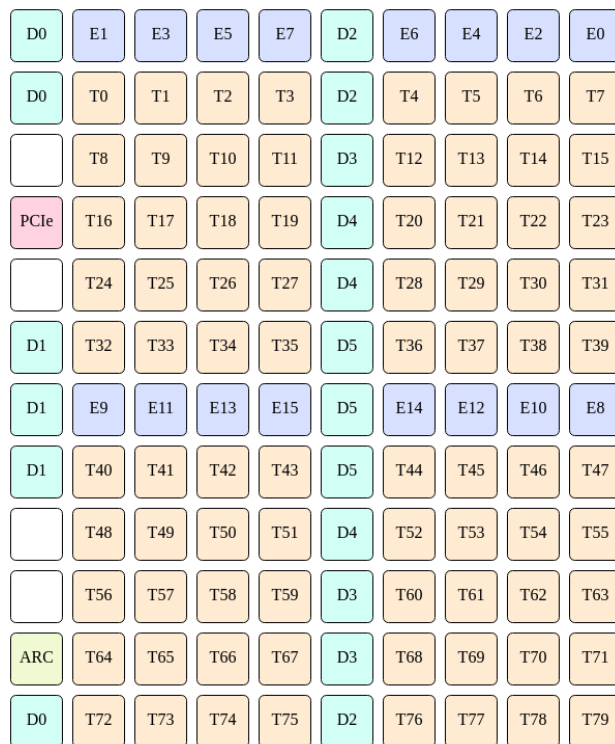
MatX

Tenstorrent

Most info obtained from the [github](#).

The market segment is flexible high-throughput compute; competing with GPUs like NVIDIA's. It has worse performance on narrow AI tasks when compared to Groq or similar narrow accelerators.

Tenstorrent is a manycore RISC-V processor using a tiled design. Here is the tile layout:



DX tiles are for DRAM access. EX tiles are for ethernet connectivity. ARC tile is for the board configuration. TX (Tensix) tiles are the compute tiles. Tensix tiles contain:

- 5x "baby" RISC-V cores
- SRAM
- NoC connections
- 1x Tensix coprocessor, which contains:
 - SRAM interface
 - 1x Matrix unit

- 1x Vector unit
- 1x Scalar unit

Opinions

- Zeb: Not a great model for our project. It has a lot of complexity, requires really good compilers, and is probably going to be worse at throughput than TPUs/Groq.

Groq

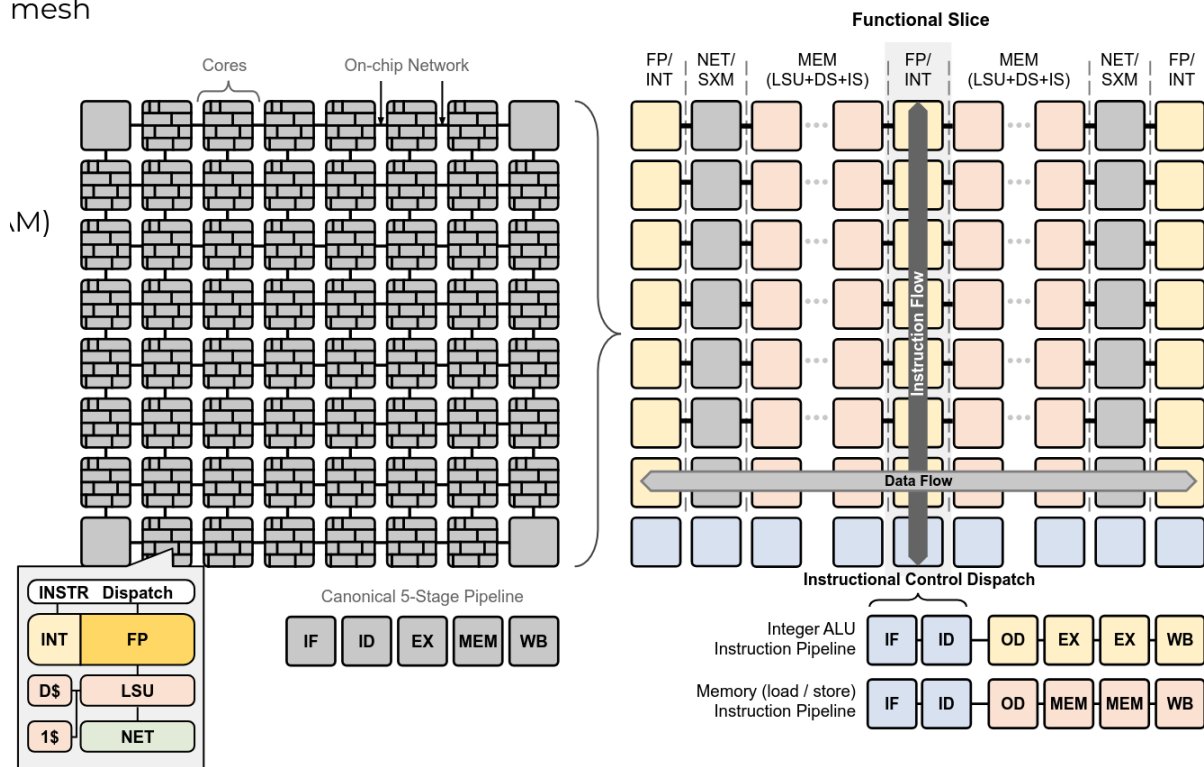
Sources:

<https://hc34.hotchips.org/assets/program/conference/day2/Machine%20Learning/HotChips34%20-%20Groq%20-%20Abts%20-%20final.pdf>

<http://pkamath.com/publications/papers/tsp-isca20.pdf>

Groq is a SIMD processor with AI-specific functional units.

mesh



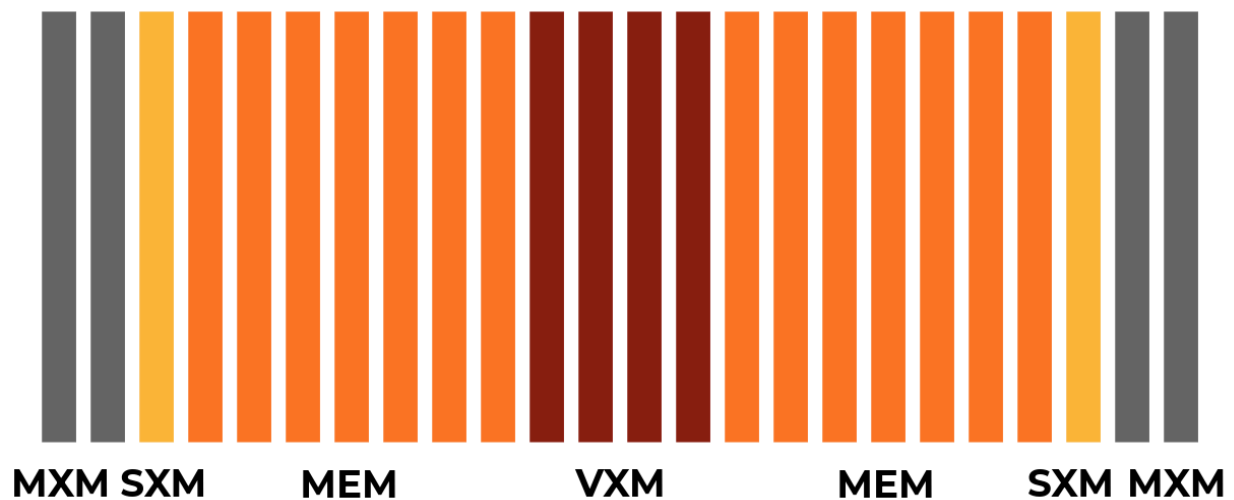
Left: conventional multicore processor

Right: Groq TSP

It amortizes the instruction control logic by sharing it across an entire column of processing elements. The processing elements are connected horizontally and have special tasks. Broadly, the PEs can be broken down into 4 types:

- MEM: has on-chip SRAM
- VXM: vector processing
 - Normal vector ops and ReLU, TanH, Exp, RSqrt
- MXM: matrix processing, supports:
 - 2x 320x320 MMs per MXM unit @ int8 (int32 accumulation)
 - 1x 160x320 MM per MXM unit @ float16 (float32 accumulation)
- SXM: data reshaping/shuffling
 - Shift up/down, Permute, Distribute, Rotate, Transpose

The PE types are arranged as such:



Notes

- All instructions execute in deterministic order and in lock-step. This means scheduling is easy to determine but must be handled by the compiler.
- The mem is *all* SRAM, which means it has deterministic timing

Maia

MTIA

Trainium / Inferentia

Taalas

Cerebras